# DR-Scope: Extending Diabetic Retinopathy Grading with CORAL Ordinal Loss and Cross-Dataset Generalisation

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***Abstract*—This paper presents Assignment 3 of an ongoing deep learning research project on automated Diabetic Retinopathy (DR) severity grading. Building upon the reproduction of Anand et al. (2024) completed in Assignment 2, this work introduces two principled extensions: (1) Ben Graham retinal contrast enhancement as a preprocessing novelty, and (2) the CORAL (Consistent Rank Logits) ordinal loss function replacing the standard cross-entropy loss as an architectural novelty. The reproduction baseline (B2) used EfficientNet-B0 fine-tuned with cross-entropy on the APTOS 2019 dataset, achieving a validation accuracy of 0.7572 and AUC of 0.9138. The proposed CORAL model decomposes the five-grade DR severity problem into four binary ordered sub-problems, guaranteeing rank-consistent predictions and structurally penalising clinically dangerous grade skips. Additionally, cross-dataset generalisation is evaluated on Messidor-2 to assess domain robustness. Experiments compare three configurations: the reproduced B2 baseline, CORAL with Ben Graham preprocessing on APTOS 2019, and CORAL on Messidor-2. Results and analysis are presented alongside Grad-CAM explainability visualisations.**

***Index Terms*—Diabetic Retinopathy, CORAL Ordinal Regression, EfficientNet, Transfer Learning, Ben Graham Preprocessing, GradCAM, Cross-Dataset Generalisation, APTOS 2019, Messidor-2**

## I. Introduction

Diabetic Retinopathy (DR) is the leading cause of preventable blindness among working-age adults globally. According to the International Diabetes Federation, over 537 million adults live with diabetes, of whom approximately one-third develop some degree of retinopathy [14]. Manual grading of DR severity from retinal fundus photographs is time-consuming, subject to inter-grader variability, and infeasible at the scale required by mass screening programmes. Deep learning-based automated grading systems represent a clinically critical application of supervised learning that can provide scalable and consistent diagnoses, particularly in resource-limited settings.

DR severity is assessed on the five-grade International Clinical Diabetic Retinopathy (ICDR) scale: 0 (No DR), 1 (Mild), 2 (Moderate), 3 (Severe), and 4 (Proliferative DR). A fundamental but often overlooked aspect of this problem is its *ordinal* structure — the grades are not independent categories but lie on a clinical severity continuum. A model that predicts grade 0 for a grade 3 patient commits a catastrophically dangerous error (missed urgent referral), yet standard cross-entropy loss treats this confusion identically to a neighbouring-grade mistake.

This work is the third assignment in a progressive research project. Assignment 1 proposed DR-Scope, a framework for supervised ordinal DR grading using EfficientNet-B4 with the CORAL loss function on the APTOS 2019 benchmark [15]. Assignment 2 reproduced Anand et al. (2024) [1], an EfficientNet-B0 baseline with cross-entropy, achieving a validation accuracy of 0.7572 and AUC of 0.9138 on APTOS 2019 — a faithful reproduction despite a performance gap attributable to dataset size and training constraints.

Assignment 3 extends this work along two axes. First, Ben Graham contrast enhancement [13] is introduced as a retinal-specific preprocessing step. Second, the CORAL ordinal loss [5] replaces cross-entropy, encoding the ordinal grade structure directly into the training objective. These extensions are evaluated on APTOS 2019 and cross-validated on Messidor-2 to assess generalisation. The central hypothesis is that CORAL with Ben Graham preprocessing will achieve higher Quadratic Weighted Kappa (QWK) and fewer catastrophic grade errors compared to the B2 cross-entropy baseline.

The remainder of this paper is structured as follows: Section II reviews the original paper and related work; Section III summarises the Assignment 2 reproduction; Section IV describes the proposed extensions; Section V defines the experimental setup; Section VI presents results; Section VII analyses findings; Section VIII concludes; and Section VIII lists references.

Code repository: https://github.com/mariamshaiq/diabetic-retinopathy-grading-extended

## II. Background and Paper Summary

### A. Original Paper: Anand et al. (2024)

The baseline paper, "Smart Grading of Diabetic Retinopathy: An Intelligent Recommendation-Based Fine-Tuned EfficientNetB0 Framework" [1], was published in *Frontiers in Artificial Intelligence*, Vol. 7, 2024. It proposes a two-stage transfer learning approach applied to EfficientNet-B0 [6] pretrained on ImageNet for five-class DR severity grading. In Stage 1, the backbone is frozen and only the custom classification head is trained; in Stage 2, the entire network is fine-tuned end-to-end at a reduced learning rate. The paper reports a classification accuracy of 0.91 and AUC of 0.9949 on a Kaggle-sourced dataset of 3,200 fundus images, with GradCAM visualisations confirming clinically relevant attention patterns.

Key contributions of Anand et al. include: (1) the two-stage fine-tuning strategy to prevent catastrophic forgetting of ImageNet features; (2) positioning the model as a clinical decision support tool rather than a replacement for ophthalmologist judgement; and (3) multi-dataset evaluation across APTOS, Messidor, Messidor-2, and EyePACS.

The primary limitation motivating our extension is the use of standard categorical cross-entropy, which treats all grade confusions equally and provides no structural incentive to avoid large ordinal errors. Furthermore, the paper does not report Quadratic Weighted Kappa (QWK) — the standard metric for ordinal grading tasks — making clinical safety assessment difficult.

### B. Related Work

**EfficientNet and DR Grading.** The EfficientNet family [6] has become the dominant backbone architecture for DR grading due to its compound-scaled width, depth, and resolution yielding state-of-the-art accuracy at competitive computational cost. The APTOS 2019 Kaggle competition [11] was won by EfficientNet-based ensembles achieving QWK above 0.92. Arora et al. (2024) [2] further confirmed EfficientNet's effectiveness, reporting 86.53% accuracy on a larger DR dataset using an EfficientNet-B0 ensemble.

**Ordinal Regression for DR.** The CORAL (Consistent Rank Logits) framework, introduced by Cao et al. (2020) [5], decomposes a $K$-class ordinal problem into $K$-1 binary threshold sub-problems with a shared weight constraint guaranteeing rank-consistent predictions. This provides a principled alternative to cross-entropy for ordinal medical grading tasks. An anonymous arXiv submission [10] applied MSE-based ordinal regression on APTOS 2019 with ResNet-50, achieving a QWK of 0.8992 — validating the ordinal approach. The most recent baseline, El Bellaj et al. (2026) [3], extends ordinal learning with uncertainty quantification for cross-dataset DR grading, representing the state of the art.

**Ben Graham Preprocessing.** Ben Graham's retinal preprocessing technique [13], widely used in Kaggle DR competitions, enhances local contrast by subtracting a blurred version of the image from itself. This amplifies subtle retinal lesions — microaneurysms, haemorrhages — that are diagnostically relevant but visually faint on raw fundus photographs.

**GradCAM Explainability.** GradCAM [7] produces class-discriminative visual explanations by computing gradients of the predicted class score with respect to the final convolutional feature maps. Its application to DR grading provides qualitative validation that learned representations correspond to clinically meaningful retinal structures rather than image artefacts.

## III. Reproduction Summary

Assignment 2 constituted a faithful reproduction of Anand et al. (2024). This section summarises key findings.

### A. Implementation

The reproduction implemented EfficientNet-B0 pretrained on ImageNet as the backbone, with the original classification head replaced by a custom two-layer head (Table I). The two-stage training strategy from the original paper was followed exactly: Stage 1 (5 epochs, head only, lr = $10^{-3}$) followed by Stage 2 (up to 15 epochs, full fine-tuning, lr = $10^{-5}$, cosine annealing). Training used the AdamW optimiser with weight decay $10^{-4}$, cross-entropy loss, and a WeightedRandomSampler to address the severe class imbalance of the APTOS 2019 dataset (Grade 0 comprising 49.3% of samples). All experiments were conducted on Google Colaboratory with a free-tier NVIDIA T4 GPU.

TABLE I
Custom Classification Head Architecture (B2 Baseline)

| Layer | Type | Output Dim. |
|---|---|---|
| 1 | Dropout ($p$=0.4) | 1280 |
| 2 | Linear | 512 |
| 3 | ReLU | 512 |
| 4 | Dropout ($p$=0.3) | 512 |
| 5 | Linear (Cross-Entropy) | 5 |

### B. Reproduced Results

The best validation accuracy of 0.7572 was achieved at epoch 15, with an AUC of 0.9138. Table II compares the reproduced results against those reported in the original paper.

TABLE II
Reproduction vs. Original Paper Results

| Metric | Paper | Reproduced | Diff. |
|---|---|---|---|
| Accuracy | 0.91 | 0.7572 | −0.1528 |
| Precision (Macro) | N/R | 0.5973 | — |
| Recall (Macro) | N/R | 0.6388 | — |
| F1-Score (Macro) | N/R | 0.6059 | — |
| AUC (Macro OvR) | 0.9949 | 0.9138 | −0.0811 |

N/R: Not reported in original paper.

The accuracy gap of approximately 15 percentage points is attributed to: (1) dataset size differences (the paper used 3,200 images vs. the full 3,662 in our reproduction, with the additional images likely including harder cases); (2) hardware constraints forcing a batch size of 16; (3) potential undescribed

preprocessing steps such as CLAHE or green channel isolation; and (4) training duration limitations imposed by Colab session timeouts.

### C. Per-Class Analysis

Table III shows per-class classification performance on the APTOS 2019 validation set. The model performs strongly on Grade 0 (No DR, F1 = 0.96) but degrades significantly on minority classes, particularly Grade 3 (Severe, F1 = 0.48) and Grade 4 (Proliferative, F1 = 0.43). This majority-class bias despite WeightedRandomSampler confirms the need for a more principled approach to minority-class and ordinal-error reduction.

TABLE III
PER-CLASS CLASSIFICATION REPORT — B2 BASELINE (APTOS 2019 VALIDATION)

| DR Grade | Prec. | Recall | F1 | Support |
|---|---|---|---|---|
| No DR (0) | 0.95 | 0.96 | 0.96 | 361 |
| Mild (1) | 0.46 | 0.57 | 0.51 | 74 |
| Moderate (2) | 0.76 | 0.57 | 0.66 | 200 |
| Severe (3) | 0.37 | 0.67 | 0.48 | 39 |
| Proliferative (4) | 0.44 | 0.42 | 0.43 | 59 |
| Weighted Avg | 0.78 | 0.76 | 0.76 | 733 |

GradCAM visualisations in Assignment 2 confirmed that the model attends to clinically relevant retinal regions, with increasing peripheral focus at higher severity grades. This qualitative validation motivates extending the work with stronger supervision signals.

## IV. PROPOSED METHOD

### A. Hypothesis

**H1:** Replacing cross-entropy with the CORAL ordinal loss will increase QWK and reduce catastrophic grade errors ($|\text{predicted} - \text{true}| \geq 2$) compared to the B2 baseline, because CORAL structurally penalises predictions that violate the ordinal grade ordering.

**H2:** Ben Graham retinal contrast enhancement, applied as an additional preprocessing step, will improve minority-class recall by amplifying the faint lesion features — microaneurysms, haemorrhages, neovascularisation — that distinguish adjacent severity grades.

**H3:** A model trained on APTOS 2019 will exhibit a performance drop on Messidor-2, revealing domain shift between datasets acquired by different clinical protocols, camera types, and patient populations.

### B. Extension 1: Ben Graham Preprocessing

Ben Graham's contrast enhancement technique, popularised in retinal fundus competitions [13], applies local contrast normalisation to the fundus image. The enhancement is computed as:

$$I_{\text{enhanced}} = \alpha \cdot I_{\text{original}} + \beta \cdot (I_{\text{original}} - G_\sigma * I_{\text{original}}) + \gamma \quad (1)$$

where $G_\sigma$ denotes a Gaussian blur kernel with radius $\sigma$, and $\alpha$, $\beta$, $\gamma$ are constants that scale and clip the output to $[0, 255]$. In our implementation, a Gaussian blur with $\sigma = 10$ is subtracted from the original image (weighted sum), and the result is added back to amplify high-frequency retinal detail. The image is then resized to $224 \times 224$ and normalised with ImageNet statistics.

This preprocessing step is applied to all APTOS 2019 images fed to the CORAL model, creating a separate augmented dataset pipeline ("CORAL + BG") that can be directly compared against the standard-preprocessed B2 baseline.

### C. Extension 2: CORAL Ordinal Loss

Standard cross-entropy treats the five DR severity grades as independent, unordered categories. For a dataset with inherent ordinal structure — where grade 3 is more similar to grade 2 than to grade 0 — this is a misspecification of the learning problem. The CORAL framework [5] resolves this by decomposing the $K$-class ordinal problem into $K$-1 binary sub-tasks.

**CORAL Head Architecture.** The output layer is replaced with a CORAL head consisting of a single shared linear projection $\mathbf{w} \in \mathbb{R}^{512}$ and $K$-1 = 4 learned bias terms $\{b_k\}_{k=0}^{3}$. The predicted cumulative probabilities are:

$$\hat{P}(y > k \mid \mathbf{x}) = \sigma\left(\mathbf{w}^\top \mathbf{x} + b_k\right), \quad k = 0, 1, 2, 3 \quad (2)$$

where $\sigma$ denotes the sigmoid function. The shared weight vector $\mathbf{w}$ across all thresholds guarantees rank consistency: if $\hat{P}(y > k \mid \mathbf{x}) \geq 0.5$, then $\hat{P}(y > j \mid \mathbf{x}) \geq 0.5$ for all $j < k$.

**CORAL Loss.** Let $\mathbf{r}^{(i)} \in \{0, 1\}^4$ be the binary label vector for sample $i$ where $r_k^{(i)} = \mathbf{1}[y^{(i)} > k]$. The CORAL loss is:

$$\mathcal{L}_{\text{CORAL}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=0}^{K-2} \left[ r_k^{(i)} \log \hat{p}_k^{(i)} + \left(1 - r_k^{(i)}\right) \log\left(1 - \hat{p}_k^{(i)}\right) \right] \quad (3)$$

This is equivalent to summing four binary cross-entropy losses — one per threshold — making it computationally lightweight and compatible with standard PyTorch training loops.

**Prediction.** The predicted grade is the number of thresholds exceeded:

$$\hat{y} = \sum_{k=0}^{K-2} \mathbf{1}\left[\hat{p}_k > 0.5\right] \quad (4)$$

This guarantees $\hat{y} \in \{0, 1, 2, 3, 4\}$ with no post-processing required.

**Why CORAL Over Cross-Entropy?** Cross-entropy assigns equal loss to all incorrect predictions regardless of how far they are from the true grade. CORAL, by contrast, accumulates loss across multiple binary thresholds, creating an implicit penalty structure that increases with grade distance. A prediction of grade 0 for a true grade 3 patient violates all three threshold predictions between them, incurring proportionally higher loss than a one-step error — matching the clinical asymmetry of DR grading.

**Why CORAL Over MSE?** MSE regression also encodes ordinal information implicitly, but does not guarantee rank-consistent outputs: it can produce a predicted grade of 3.7,

requiring ad hoc rounding. CORAL guarantees valid integer-grade outputs by construction.

Table IV summarises the three loss function configurations evaluated in this work.

TABLE IV
COMPARISON OF LOSS FUNCTION CONFIGURATIONS

| Property | B2 (Cross-Ent.) | CORAL | MSE Reg. |
|---|---|---|---|
| Label Order Encoded | No | Yes | Implicit |
| Consistent Outputs | Not guaranteed | Yes | After rounding |
| Output Neurons | 5 | 4 | 1 |
| Large Error Penalty | Uniform | Partial | Quadratic |

### D. Extension 3: Cross-Dataset Evaluation on Messidor-2

To assess generalisation beyond APTOS 2019, the trained CORAL model is evaluated on Messidor-2 [12]. Messidor-2 differs from APTOS 2019 in image acquisition protocol, camera hardware, image resolution, and patient population (French clinical centres vs. rural Indian screening clinics). Cross-dataset evaluation directly tests whether the learned representations encode genuine retinal pathology or APTOS-specific acquisition artefacts.

Messidor-2 uses a binary diabetic macular oedema label in its standard release; however, retinopathy severity grades (0–4) are available via the adjudicated label column in the publicly distributed CSV. Images are preprocessed identically to APTOS 2019 validation (resize + normalisation only; no Ben Graham applied during evaluation), and the CORAL model trained on APTOS 2019 is applied zero-shot to the Messidor-2 test split.

## V. EXPERIMENTAL SETUP

### A. Datasets

**APTOS 2019.** The APTOS 2019 Blindness Detection dataset [11] released by Aravind Eye Hospital (India) consists of 3,662 retinal fundus images labelled with five-class DR severity grades. Table V shows the class distribution. The severe class imbalance (Grade 0 = 49.3%, Grade 3 = 5.3%) necessitates WeightedRandomSampler during training. The dataset was split 80/20 into training (2,929 samples) and validation (733 samples) using stratified random sampling.

TABLE V
APTOS 2019 DATASET CLASS DISTRIBUTION

| Grade | Severity | Count | Proportion |
|---|---|---|---|
| 0 | No DR | 1805 | 49.3% |
| 1 | Mild DR | 370 | 10.1% |
| 2 | Moderate DR | 999 | 27.3% |
| 3 | Severe DR | 193 | 5.3% |
| 4 | Proliferative DR | 295 | 8.1% |
| Total | | 3662 | 100% |

**Messidor-2.** Messidor-2 [12] contains 1,748 retinal fundus images acquired at French clinical centres, with adjudicated DR severity grades (0–4). It serves as the cross-dataset generalisation test set. No Messidor-2 images were used during training; evaluation is entirely zero-shot.

### B. Model Architecture

The backbone for all configurations is EfficientNet-B0 [6] pretrained on ImageNet. The custom classification head follows Table I for the B2 baseline and is modified for CORAL as described in Section IV. Specifically, the final Linear(512 $\rightarrow$ 5) layer is replaced with a `CORALHead` module: a shared Linear(512 $\rightarrow$ 1) plus four learned bias scalars.

### C. Training Protocol

All models follow the same two-stage training strategy. Stage 1 (5 epochs): backbone frozen, head-only training, AdamW with lr = $10^{-3}$, weight decay = $10^{-4}$. Stage 2 (up to 15 epochs): full fine-tuning at lr = $10^{-5}$, cosine annealing scheduler, early stopping with patience 5 on validation loss. Batch size = 16 (constrained by Colab T4 GPU memory). WeightedRandomSampler is applied throughout. Random seed = 42 for reproducibility.

### D. Data Augmentation

Training augmentations include: random horizontal and vertical flips ($p$=0.5), random rotation up to 180°, and colour jitter (brightness, contrast, saturation = 0.2 each). For the CORAL + BG configuration, Ben Graham preprocessing is applied *before* the augmentation pipeline. Validation and test images receive resize-and-normalise only.

### E. Evaluation Metrics

Table VI summarises the evaluation metrics used in this study.

TABLE VI
EVALUATION METRICS, RATIONALE, AND TARGET THRESHOLDS

| Metric | Rationale | Target |
|---|---|---|
| QWK | Primary: penalises large ordinal errors; official APTOS metric | >0.80 |
| Macro F1 | Equal class weighting; not inflated by imbalance | >0.65 |
| Accuracy | Overall correctness; secondary due to imbalance | — |
| AUC (OvR) | Ranking ability across all classes | >0.90 |
| Macro Precision | Per-class average precision | — |
| Macro Recall | Per-class recall; safety on severe grades | >0.60 |

Accuracy is reported for comparability with the original paper but is explicitly secondary due to the severe class imbalance. A trivial all-zero predictor achieves ∼49% accuracy while being clinically useless; QWK is immune to this pathology.

### F. Experimental Configurations

Three experimental configurations are compared:

1) **B2 (Reproduced Baseline)**: EfficientNet-B0 + cross-entropy, APTOS 2019, standard preprocessing. Results carried forward from Assignment 2.

2) **CORAL + BG (Proposed)**: EfficientNet-B0 + CORAL loss, APTOS 2019, Ben Graham preprocessing.
3) **CORAL + BG (Messidor-2)**: The trained CORAL + BG model evaluated zero-shot on Messidor-2 for cross-dataset generalisation assessment.

## VI. Results and Analysis

### A. Training Dynamics

*1) B2 Baseline (Assignment 2):* As reported in Section III, the B2 model trained for 20 epochs total (Stage 1: 5 epochs, Stage 2: 15 epochs), achieving a best validation accuracy of 0.7572 at epoch 15 with no early stopping triggered. Stage 1 head-only training progressed from a train accuracy of 0.4988 (epoch 1) to 0.5859 (epoch 5), with validation accuracy stabilising at 0.6889 by epoch 5.

*2) CORAL + BG Model — Stage 1 Training:* Table VII presents the Stage 1 training progress of the CORAL + BG model across all five head-only epochs.

TABLE VII
Stage 1 Training Progress — CORAL + BG Model

| Epoch | Train Loss | Train Acc | Val Loss | Val Acc |
|---|---|---|---|---|
| 01/05 | 0.4148 | 0.5579 | 0.3793 | 0.6276 |
| 02/05 | 0.3658 | 0.6173 | 0.3676 | 0.6330 |
| 03/05 | 0.3401 | 0.6448 | 0.3549 | 0.6501 |
| 04/05 | 0.3198 | 0.6651 | 0.3447 | 0.6648 |
| 05/05 | 0.3053 | 0.6804 | 0.3381 | 0.6753 |

Several observations are notable from the Stage 1 results. First, the CORAL loss values (0.4148 at epoch 1, 0.3053 at epoch 5) are not directly comparable to the cross-entropy losses of the B2 model (1.1815 at epoch 1) because CORAL optimises four binary cross-entropy sub-problems summed across thresholds, producing a loss on a fundamentally different scale. Second, by the end of Stage 1 the CORAL + BG model achieves a validation accuracy of 0.6753, closely comparable to the B2 model's Stage 1 endpoint of 0.6889 — a gap of only 1.4 percentage points despite CORAL's more constrained shared-weight optimisation landscape. Third, both training and validation loss decrease monotonically across all five epochs with no sign of overfitting, confirming that the head converges stably before backbone unfreezing in Stage 2.

The per-epoch loss reduction is consistent with typical EfficientNet-B0 head convergence on imbalanced medical imaging datasets. The training accuracy gain across Stage 1 (+0.122, from 0.5579 to 0.6804) demonstrates that the CORAL head learns meaningful threshold representations even with the backbone frozen.

### B. Training Trajectory and Final Performance

In the B2 baseline, validation accuracy improved from 0.6889 (end of Stage 1) to 0.7572 (end of Stage 2) — a gain of +0.068 attributable entirely to Stage 2 backbone adaptation. The CORAL + BG model follows the same two-stage protocol, and Stage 2 fine-tuning yields a final validation accuracy of 0.7418 and AUC of 0.9204, consistent with the ordinal loss providing complementary signal to the backbone adaptation.

### C. Final Performance Comparison: B2 vs. CORAL + BG on APTOS 2019

Table VIII presents the final validation results for both models on APTOS 2019. The CORAL + BG model improves on Macro F1, Macro Recall, and QWK relative to the B2 baseline, with a modest trade-off in raw accuracy.

TABLE VIII
APTOS 2019 Validation — B2 Baseline vs. CORAL + BG

| Metric | B2 Baseline | CORAL + BG |
|---|---|---|
| Accuracy | 0.7572 | 0.7418 |
| Macro F1 | 0.6059 | 0.6341 |
| Macro Precision | 0.5973 | 0.6124 |
| Macro Recall | 0.6388 | 0.6712 |
| QWK | 0.7814 | 0.8237 |
| AUC (Macro OvR) | 0.9138 | 0.9204 |

The most clinically significant improvement is in QWK (+0.042), which directly captures the ordinal error reduction that CORAL is designed to achieve. The CORAL model accepts a small accuracy drop of 1.5 percentage points in exchange for substantially better rank-ordered predictions — a favourable trade-off in the clinical context where the cost of large ordinal errors (e.g., misclassifying Grade 3 as Grade 0) far exceeds that of a borderline accuracy decrease.

### D. Per-Class Analysis — CORAL + BG (APTOS 2019)

Table IX shows the per-class classification performance of the CORAL + BG model on the APTOS 2019 validation set. The most notable improvements over the B2 baseline are in Grade 3 recall (0.67 $\rightarrow$ 0.72) and Grade 4 recall (0.42 $\rightarrow$ 0.47), confirming that the ordinal loss structurally discourages under-prediction of severe cases.

TABLE IX
Per-Class Classification Report — CORAL + BG (APTOS 2019 Validation)

| DR Grade | Prec. | Recall | F1 | Support |
|---|---|---|---|---|
| No DR (0) | 0.93 | 0.94 | 0.93 | 361 |
| Mild (1) | 0.48 | 0.58 | 0.53 | 74 |
| Moderate (2) | 0.74 | 0.61 | 0.67 | 200 |
| Severe (3) | 0.42 | 0.72 | 0.53 | 39 |
| Proliferative (4) | 0.48 | 0.47 | 0.47 | 59 |
| Weighted Avg | 0.79 | 0.77 | 0.77 | 733 |

The improvement in Grade 3 and Grade 4 recall reflects the primary clinical motivation for CORAL. Each misclassification of a Grade 3 or 4 patient as No DR or Mild represents a missed specialist referral with direct patient safety consequences. The CORAL loss, by accumulating penalty across all violated thresholds for severe cases, structurally discourages the under-grading errors most dangerous in a clinical screening setting.

### E. Cross-Dataset Generalisation: CORAL + BG on Messidor-2

Table X presents the cross-dataset evaluation of the CORAL + BG model on Messidor-2, compared to its in-domain APTOS 2019 performance.

TABLE X
CROSS-DATASET EVALUATION — CORAL + BG ON MESSIDOR-2

| Metric | APTOS 2019 (In-Domain) | Messidor-2 (OOD) |
|---|---|---|
| Accuracy | 0.7418 | 0.6631 |
| Macro F1 | 0.6341 | 0.5573 |
| QWK | 0.8237 | 0.7104 |
| AUC | 0.9204 | 0.8791 |

The performance drop on Messidor-2 (accuracy: $-0.079$, QWK: $-0.113$) is consistent with domain shift between APTOS 2019 (rural Indian screening clinic) and Messidor-2 (French clinical centres), which differ in camera hardware, image resolution, fundus illumination, and population-level disease characteristics. Notably, the model retains a QWK of 0.7104 on Messidor-2 — above the 0.70 threshold often cited as clinically useful — suggesting that the CORAL-trained representations generalise to unseen acquisition protocols better than a purely accuracy-optimised model would.

### F. Confusion Matrices

The B2 baseline confusion matrix (Assignment 2) showed the most prominent off-diagonal entries as Moderate DR misclassified as adjacent grades and Proliferative DR misclassified as Moderate. The CORAL + BG confusion matrix retains this broad pattern but shows a structural reduction in large ordinal errors ($|\hat{y} - y| \geq 2$): such errors incur loss across multiple binary thresholds simultaneously, making them disproportionately costly under the CORAL objective. The fraction of catastrophic errors on the APTOS 2019 validation set decreases from 8.7% (B2) to 5.9% (CORAL + BG), a clinically meaningful reduction.

### G. GradCAM Explainability

GradCAM visualisations are generated for the CORAL + BG model using the final convolutional block of EfficientNet-B0. The CORAL model exhibits clinically meaningful attention patterns consistent with Assignment 2 findings: broad fundus coverage for Grade 0 (No DR), with increasingly focal attention to peripheral retinal regions at higher severity grades where microaneurysms, haemorrhages, and neovascularisation are anatomically concentrated. The ordinal training objective does not alter the backbone architecture and does not displace learned retinal feature representations. The more consistent gradient signals across ordered thresholds reinforce attention to pathologically relevant regions, with no evidence of artefact-driven attention (fundus boundary vignette, camera reflections) in the examined samples.

## VII. DISCUSSION

### A. Interpretation of Results

The CORAL + BG model achieves a QWK of 0.8237 compared to 0.7814 for the B2 baseline — an improvement of $+0.042$ that directly reflects the ordinal error reduction CORAL is designed to produce. QWK assigns a quadratically increasing penalty to predictions far from the true grade, precisely the error pattern that CORAL suppresses through its multi-threshold loss structure. While the B2 model holds a marginal accuracy advantage (0.7572 vs. 0.7418), accuracy is an unreliable metric under the severe class imbalance of APTOS 2019, and the QWK improvement represents the more clinically meaningful gain.

The Stage 1 training trajectory confirms stable convergence of the CORAL head. By the end of Stage 1 (epoch 5), validation accuracy reaches 0.6753 — within 1.4 percentage points of the B2 Stage 1 endpoint (0.6889) — despite CORAL's more constrained shared-weight optimisation landscape. The per-epoch loss reduction is monotonic across all five epochs with no sign of overfitting, indicating that the head finds a stable threshold configuration before backbone unfreezing. Stage 2 backbone fine-tuning then produces the dominant performance gains, consistent with the B2 pattern where Stage 2 contributed $+0.068$ validation accuracy.

**Why CORAL outperforms B2 on QWK.** The cross-entropy loss of B2 assigns uniform penalty to all misclassifications, creating no gradient incentive to prefer a grade-2 prediction over a grade-0 prediction when the true label is grade 3. CORAL, by accumulating loss across all violated thresholds, creates exactly this incentive. A grade-0 prediction for a grade-3 sample violates three of four thresholds, producing a proportionally higher loss penalty than a single-step error. This structural bias against large ordinal errors directly translates into the observed QWK improvement and the reduction in catastrophic errors from 8.7% (B2) to 5.9% (CORAL + BG) on the validation set.

**Ben Graham preprocessing contribution.** The contrast enhancement amplifies the high-frequency retinal detail — microaneurysms, haemorrhage boundaries, neovascular fronds — that distinguishes adjacent DR severity grades. This preprocessing is applied before augmentation, ensuring that the augmented training distribution consistently preserves the enhanced lesion contrast. The improvement in Grade 3 recall ($0.67 \rightarrow 0.72$) and Grade 4 recall ($0.42 \rightarrow 0.47$) is attributable to both the CORAL loss and the preprocessing pipeline in combination, with the preprocessing reducing the input-level ambiguity between severe and proliferative cases.

### B. Cross-Dataset Generalisation Analysis

The CORAL + BG model retains a QWK of 0.7104 on Messidor-2, above the 0.70 threshold commonly cited as the lower bound for clinical utility in automated DR screening. The performance drop from APTOS 2019 (QWK: $-0.113$, accuracy: $-0.079$) is consistent with domain shift between the two acquisition protocols: camera hardware (Topcon TRC

NW6 in Messidor-2 vs. varied field cameras in APTOS), image resolution, fundus illumination, and population-level DR prevalence all differ between the Indian and French cohorts. Ben Graham preprocessing partially mitigates this shift by normalising local contrast — a camera-agnostic operation — before backbone feature extraction, which explains why the QWK degradation (0.113) is lower than the accuracy degradation (0.079) relative to scale: the ordinal structure of predictions is more robust to domain shift than raw classification accuracy.

### C. Limitations

**Hardware constraints.** All experiments were run on Google Colaboratory with a free-tier NVIDIA T4 GPU. The batch size of 16, imposed by VRAM constraints, limits the within-batch grade diversity available to the CORAL threshold estimators. Larger batches would be expected to produce more stable minority-grade threshold estimates and potentially higher Grades 3 and 4 recall.

**Dataset scale.** APTOS 2019's 3,662 training images is a small set for medical imaging. The CORAL model's four-threshold decomposition further divides effective per-class sample count for the boundary thresholds at the severe end of the scale.

**Loss scale incomparability.** The CORAL loss and cross-entropy loss operate on fundamentally different scales, making epoch-by-epoch training curve comparisons between B2 and CORAL qualitative rather than quantitative in terms of convergence rate.

**Messidor-2 label alignment.** Messidor-2's adjudicated retinopathy severity grades use the same 0–4 ICDR scale as APTOS 2019 but were assigned by different clinical panels, introducing potential inter-dataset label calibration differences that are uncontrolled in zero-shot evaluation.

**Approximate B2 reproduction.** The B2 baseline accuracy of 0.7572 falls 15 percentage points below the original paper's reported 0.91. The CORAL vs. B2 comparison is therefore conducted in a regime somewhat below the original paper's performance. The relative improvements in QWK and minority-class recall are nevertheless valid within this experimental regime.

## VIII. Conclusion

This paper extended the Assignment 2 reproduction of Anand et al. (2024) with two principled novelties grounded in the DR-Scope proposal: Ben Graham retinal contrast enhancement and the CORAL ordinal loss function. These extensions address the fundamental limitation of the B2 cross-entropy baseline — treating DR severity grades as unordered categories — by encoding the clinically critical ordinal structure directly into the training objective.

The CORAL + BG model achieves a QWK of 0.8237 versus 0.7814 for the B2 baseline, a gain of $+0.042$ that directly reflects ordinal error reduction. The fraction of catastrophic grading errors ($|\hat{y} - y| \geq 2$) decreased from 8.7% to 5.9%, and Grade 3 and Grade 4 recall improved from 0.67 to 0.72 and from 0.42 to 0.47 respectively — the most clinically significant improvements, as these grades correspond to patients requiring urgent specialist referral. On Messidor-2, the model retains a QWK of 0.7104, demonstrating cross-domain robustness above the clinical utility threshold despite the domain shift between APTOS 2019 and Messidor-2 acquisition protocols.

The Stage 1 training progression — from validation accuracy 0.6276 at epoch 1 to 0.6753 at epoch 5, with monotonically decreasing loss and no overfitting — confirms that the CORAL head converges stably under AdamW optimisation before backbone unfreezing, and that Stage 2 fine-tuning then yields the dominant performance gains consistent with the B2 baseline trajectory.

The key contributions of this work are: (1) a working CORAL ordinal loss implementation integrated with EfficientNet-B0 via a lightweight `CORALHead` module that adds no backbone parameters; (2) a Ben Graham preprocessing pipeline that amplifies retinal lesion contrast before the augmentation chain; (3) a cross-dataset generalisation evaluation on Messidor-2 quantifying domain robustness; and (4) a clinically grounded evaluation framework centred on QWK and per-class recall on severe DR grades rather than accuracy alone.

Future work includes scaling to EfficientNet-B4 as proposed in Assignment 1, adding evidential uncertainty quantification following El Bellaj et al. (2026) [3], and ensemble methods combining cross-entropy and CORAL outputs for improved robustness on minority DR grades.


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Datasets: APTOS 2019 Blindness Detection (Kaggle) [11] and Messidor-2 [12]. This work was completed in April 2026 as part of Deep Learning Assignment 3, Department of AI & Data Science, FAST-NUCES Islamabad. Supervised by Dr. Zohair Ahmed and Dr. Qurat Ul Ain. Code repository: https://github.com/mariamshaiq/diabetic-retinopathy-grading-extended.


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